HOMEWORK -2

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1) "predictable no fun"

Priors $P(-) = 3/5$ $P(+) = \frac{2}{5}$

$P(\text{predictable}|-) = \frac{2+1}{14+20} = \frac{3}{34}$

$P(\text{no}|-) = \frac{1+1}{20+14} = \frac{2}{34}$

$P(\text{fun}|-) = \frac{0+1}{34}$

$P(\text{Predictible}|+) = \frac{0+1}{20+9} = \frac{1}{29}$

$P(\text{no}|-) = \frac{0+1}{29} = \frac{1}{29}$

$P(\text{fun}|+) = \frac{1+1}{29} = \frac{2}{29}$

$P(d|-) = \frac{3}{5} \times \frac{3}{34} \times \frac{2}{34} \times \frac{1}{34}$

$= 9.163 \times 10^{-5}$ $>$

$P(d|+) = \frac{2}{5} \times \frac{1}{29} \times \frac{1}{29} \times \frac{2}{29}$

$= 3.294 \times 10^{-5}$

prediction:- Negative class.

2) (i) when a model reinforces stereotypes. Kiritchenko & Mohammad (2018) showed sentiment models rated some demographic groups more negatively.

(ii) Toxicity systems may unfairly silence marginalized groups by

(iii) models are trained mostly on standard english not other dialects.

3) Per class precision and recall

$Prec = \frac{TP}{TP+FP}$ $Recall = \frac{TP}{TP+FN}$

(i)

Cat:-

$Prec = \frac{5}{5+15} = 0.25$

$Rec = \frac{5}{5+15} = 0.25$

Dog:-

$Prec = \frac{20}{10+20+15} = 0.44$

$Rec = \frac{20}{20+10+15} = 0.44$

Rabbit

$Prec = \frac{10}{5+10+10} = \frac{10}{25} = 0.4$

$Rec = \frac{10}{5+10+10} = 0.4$

(ii) macro averaged

$Precision = \frac{0.25+0.44+0.4}{3} \approx 0.367$

$Recall = \frac{0.25+0.44+0.4}{3} \approx 0.367$

Treats all classes equally

Good when class imbalance exists

Micro averaged

Total TP = 5+20+10 = 35

Total predicted = 90

micro precision = $\frac{35}{90} = 0.3889$

micro recall = $\frac{35}{90} = 0.3889$

Aggregates across all classes

Favours performance on dominant classes.

4) c(<s>) = 3, c(I) = 2, c(love) = 2, c(deep) = 2, c(learning) = 2 c(NLP) = 1.

1. Bigram sentence probabilities.

(a) S1 : <s> I love NLP </s>

$p(I|<s>) = \frac{2}{3}$

$p(love|I) = \frac{2}{2} = 1$

$p(NLP|love) = \frac{1}{(1+1)} = 1/2$.

$p(</s>|NLP) = 1/1 = 1$

$p(S1) = 2/3 \times 1 \times 1/2 \times 1 = 1/3 \approx 0.33$

(b) S_2 : <s> I love deep learning </s>

$P(I \mid <s>) = \frac{2}{3}$

$P(love \mid I) = 1$

$P(deep \mid love) = 1/2$

$P(learning \mid deep) = 2/2 = 1$

$P(</s> \mid learning) = 1/1+1 = 2$

$P(S_2) = 2/3 \times 1 \times 1/2 \times 1 \times 1/2 = 0.167$.

c) $P(S_1) > P(S_2)$ under bigram model.

2) Zero probability problem : $P(noodle \mid ate)$

a) count(noodle after ate) = 0

Total count of ate = 12 (as there are words after ate)

$$P(noodle \mid ate) = \frac{0}{12} = 0$$

b) zero probability makes sentence scoring and perplexity unreliable

to overcome this we use +1 smoothing.

c) Vocabulary size = 10

bigrams after "ate" = 12

$$P(noodle \mid ate) = \frac{0+1}{12+10} = \frac{1}{22} \approx 0.045 \rightarrow \text{Some probability}$$

5) $P(cats \mid I, like) = \frac{C(I\ like\ cats)}{C(I\ like}$

1)

$$= \frac{1}{2}$$

2) P(dogs | you, like):

no trigram "you like dogs" exists. trigram probability = 0.

So, backoff diagram.

$c(like, dogs) = 1$

$c(like) = 3$

$P(dogs | like) = 1/3$.

3. when trigram prob = 0, we use bigram (backoff) to avoid zero